Experiment 13

Problem Statement:-

Write a Python program using modular programming to generate the Fibonacci sequence up to a given maximum value.

Algorithm:-

Generate Fibonacci Sequence up to a Limit

1. Start
2. Input a number `n`
3. Check if `n` is greater than 0

- If **no**, print *"Please enter a positive integer"* and stop
- If **yes**, continue

4. Call a function to generate the Fibonacci sequence up to `n`

generate_fibonacci_sequence(max_value)

5. If `max_value ≤ 0`

- Return an empty list

6. Initialize:

- `a = 0` (first seed)
- `b = 1` (second seed)
- `seq = []` (empty list to store sequence)

7. While `a ≤ max_value`:

- Add `a` to `seq`
- Update values:
  - `a = b`
  - `b = a + b` (use previous values)

8. Return `seq`
9. Store returned sequence in `fib_series`
10. Convert each number to string
11. Join them with spaces
12. Print the result
13. End